Literature Review: Constraints on Vector Portal, Higgs Portal, $U(1)_{B-L}$, $U(1)_{L_i - L_j}$, and $U(1)_B$ Dark Matter Portals

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Contents

1 Introduction

This document collects the most up-to-date constraints—from direct detection, colliders, astrophysics, and cosmology—on dark matter (DM) models built on five portal frameworks:

(i) **Vector portal** (dark photon via kinetic mixing),

(ii) **Higgs portal** (singlet scalar mixing with the SM Higgs),

(iii) $U(1)_{B-L}$ (baryon minus lepton number),

(iv) $U(1)_{L_i - L_j}$ with $(i, j) = (e, \mu),\ (e, \tau),\ (\mu, \tau)$,

(v) $U(1)_B$ (gauged baryon number / baryon portal).

The first two are the most minimal and widely studied portals; the latter three are anomaly-free $U(1)$ gauge extensions with a massive Z'. The phenomenology differs dramatically depending on whether the mediator couples to quarks, leptons, or both at tree level.

2 Vector Portal (Dark Photon)

A new $U(1)_D$ gauge boson A' (dark photon) kinetically mixes with hypercharge, $\mathcal{L} \supset -\frac{\varepsilon}{2} F'_{\mu\nu} B^{\mu\nu}$, giving it an effective coupling εe to all electrically charged SM particles. DM (Dirac fermion χ or complex scalar) is charged under $U(1)_D$ with gauge coupling g_D. This is the most minimal portal and the benchmark for the sub-GeV DM programme.

2.1 Direct detection

- **LZ / XENONnT / PandaX (2024–2025):** For GeV-scale DM, SI scattering via t-channel A' exchange gives $\sigma_{\rm SI} \propto \varepsilon^2 g_D^2 / m_{A'}^4$. The LZ 2025 result ($\sigma_{\rm SI} \sim 9.2 \times 10^{-48}$ cm^2 at 30 GeV) rules out nearly all thermal-relic scenarios with $m_{A'} \lesssim 1$ TeV unless ε is extremely small [1, 27].
- **Sub-GeV DM–electron scattering:** DAMIC-M (2025) [17], SENSEI, and DarkSide-50 set world-leading limits on DM–electron scattering for $m_{\rm DM} \sim 1$ MeV–1 GeV. These directly probe the ε–$m_{A'}$ plane and exclude large regions of hidden-sector DM parameter space.
- **Ultralight dark photon DM (A' is the DM):** For $m_{A'} \sim 10^{-22}$–1 eV, absorption in direct-detection experiments (ADMX, DM-Radio, BREAD, SuperCDMS) constrains ε via the kinetic-mixing–induced electric field oscillation [28].

2.2 Collider and fixed-target constraints

- **Beam dumps and fixed-target experiments:** E137, E141, E774, νCal, NA64-e (visible and invisible modes), and the planned LDMX provide the leading bounds in the ε–$m_{A'}$ plane for $m_{A'} \sim 1$ MeV–1 GeV. NA64-e invisible-mode results exclude $\varepsilon \gtrsim$ few $\times 10^{-4}$ for $m_{A'} \lesssim 200$ MeV. LDMX is projected to cover the thermal-relic target for all sub-GeV DM masses [33].
- **BaBar mono-photon:** Excludes $\varepsilon \gtrsim 10^{-3}$ for $m_{A'} \lesssim 8$ GeV via $e^+e^- \to \gamma A'(\to$ invisible).
- **LHC:** For $m_{A'} \gtrsim 10$ GeV, Drell-Yan dilepton searches and displaced-vertex signatures constrain ε; combined constraints compiled in Ref. [30].
- **Heavy-ion collisions:** Thermal A' production in the QGP provides sensitivity in the MeV mass range via dilepton spectra [32].

2.3 Cosmological and astrophysical constraints

- **BBN / N_{eff}:** A thermalised A' that decays into e^+e^- or neutrinos after neutrino decoupling shifts N_{eff} and alters light-element abundances. Joint CMB+BBN data exclude $m_{\text{DM}} \lesssim 4$ MeV at 95% CL for all DM spins and A' masses [34].

- **CMB spectral distortions:** Dark photon oscillations ($A' \leftrightarrow \gamma$) inject or remove photons from the CMB bath; COBE/FIRAS constrains ε for $m_{A'} \sim 10^{-14}$–10^{-4} eV, with recent work showing previous bounds were underestimated [35].

- **CMB anisotropies:** For s-wave annihilation ($\chi\bar{\chi} \to A'A'$ or $\to e^+e^-$), energy injection at recombination is constrained by Planck. This excludes complex scalar and Majorana DM in the secluded annihilation channel for $m_{\text{DM}} \lesssim 10$ GeV [34].

- **Stellar cooling:** Horizontal-branch stars, red giants, and the Sun constrain $\varepsilon \lesssim 10^{-10}$–$10^{-14}$ for $m_{A'} \lesssim$ keV via Compton-like and bremsstrahlung production.

- **SN1987A:** A' production in the proto-neutron star provides the strongest astrophysical bound for $m_{A'} \sim 1$–300 MeV, recently updated with improved shock-cooling modelling [14].

- **Dark photon DM production mechanisms:** In models where A' *is* the DM, early-universe production via misalignment or inflationary fluctuations is viable, but dark Higgs dynamics can collapse the dark sector into cosmic strings, restricting the viable parameter space [28].

2.4 Key references

- "The Heavy Dark Photon Handbook" [29]: comprehensive compilation of cosmological and astrophysical bounds (SN1987A, failed SNe, BBN, CMB distortions, γ-ray limits) for MeV-scale dark photons. *The single best review reference for this portal.*

- "Combined constraints on dark photons" [30]: collider, cosmology, and astrophysics combined in one plot.

- "GeV-scale thermal DM from dark photons: tightly constrained, yet allowed" [31]: the most recent (2025) comprehensive analysis of the full interplay between relic density, direct detection (LZ), indirect detection, and collider bounds for the dark Abelian Higgs model with Dirac DM.

- "WIMP DM within the dark photon portal" [27]: Dirac and scalar DM up to 1 TeV; derives lower limits on dark couplings from relic density and compares with LZ SI bounds.

3 Higgs Portal

A real singlet scalar S couples to the SM via the renormalisable operator $\lambda_{HS}\, S^2\, |H|^2$. After electroweak symmetry breaking, S mixes with the physical Higgs boson h with mixing angle θ, inheriting all SM Higgs couplings suppressed by $\sin\theta$. This is the simplest UV-complete DM portal if S is stabilised by a $\mathbb{Z}_2$ symmetry ($S \to -S$), or, more generally, if S is the mediator to a dark sector (as in the secluded DM framework studied in this project).

3.1 Direct detection

- **LZ + XENONnT + PandaX (2024–2025):** SI scattering proceeds via t-channel h/S exchange with $\sigma_{\text{SI}} \propto \sin^2\theta\, f_N^2/m_S^4$. LZ 2025 limits leave only a narrow viable window near the Higgs resonance, $m_S \in [60.5, 62.5]$ GeV, with the Higgs-portal coupling constrained to $\lambda_{HS} \sim (1.7$–$4.7) \times 10^{-4}$ [36].

- **Sub-GeV regime:** For light mediators ($m_S \lesssim 5$ GeV), DM–nucleon scattering is enhanced at low momentum transfer. CRESST-III and NEWS-G provide the leading bounds for $m_{\rm DM} \sim 0.1$–1 GeV. Cosmic-ray–accelerated DM (CRDM) scattering off nucleons extends sensitivity to even lighter DM [37].

- **Light self-interacting DM:** If S mediates DM self-interactions (as in secluded models), the combination of direct detection + BBN + relic density constrains the mediator mass and coupling jointly [37].

3.2 Collider constraints

- **Invisible Higgs width:** $h \to SS$ (if $m_S < m_h/2$) contributes to the invisible branching ratio. ATLAS Run 2 constrains $\mathrm{BR}(h \to \mathrm{inv}) < 0.107$ at 95% CL, directly bounding λ_{HS} [36].

- **Off-shell / heavy scalar searches:** For $m_S > m_h/2$, LHC searches for heavy scalars decaying to $WW/ZZ/hh$ constrain $\sin\theta$ as a function of m_S. Vector-boson fusion and mono-Z channels probe freeze-in scenarios at the HL-LHC [40].

- **Higgs signal-strength measurements:** All SM Higgs couplings are universally reduced by $\cos\theta$; LHC combined signal-strength fits constrain $\sin^2\theta \lesssim 0.05$–$0.12$ depending on the analysis [38].

3.3 Indirect detection

- **Fermi-LAT dwarf spheroidals:** 14 years of data from 16 dwarfs constrain the DM annihilation cross section for $m_{\rm DM} \sim 10$ GeV–100 TeV. For Higgs-portal DM annihilating to $b\bar{b}$, the thermal cross section is excluded below $m_{\rm DM} \sim 100$ GeV except near the h-funnel [43].

- **Gamma-ray lines (Fermi-LAT, DAMPE):** 15 years of Fermi-LAT data show no spectral lines; the first coupled $\gamma\gamma + Z\gamma$ double-line analysis constrains the loop-induced $\chi\chi \to \gamma\gamma$ rate, which is relevant for Higgs-portal models at $m_{\rm DM} \sim m_h/2$ [44].

- **CMB (Planck):** Energy injection from s-wave annihilation at recombination provides model-independent bounds; for p-wave–dominated Higgs-portal scenarios these are weaker [41].

3.4 Cosmological and astrophysical constraints

- **Relic density:** The combination of s-channel h-exchange and (if $m_{\rm DM} > m_h$) t-channel annihilation to hh determines the thermal relic abundance. Global fits (GAMBIT) find that only the resonance region $m_{\rm DM} \approx m_h/2$ and the high-mass regime $m_{\rm DM} \gtrsim 500$ GeV survive all constraints [39].

- **Electroweak phase transition:** The singlet scalar can catalyse a strong first-order EWPT (relevant for baryogenesis), linking Higgs-portal DM to gravitational-wave signals at LISA. Recent work connects this to the singlet–fermion DM Higgs-portal framework [42].

- **BBN:** For secluded models where the mediator S is long-lived, late decays to SM particles ($b\bar{b}$, $\tau^+\tau^-$, etc.) can disrupt BBN. Constraints depend on the S lifetime and branching ratios—directly tied to the decay widths computed in our `phi_decay` module.

3.5 Key references

- "Dark Matter through the Higgs portal" [38]: the most comprehensive review of Higgs-portal DM, covering scalar/fermionic/vector DM, EFT vs. UV-complete models, and all constraint classes.
- GAMBIT global analysis [39]: state-of-the-art global fit including relic density, direct/indirect detection, and collider constraints.
- "New constraints on singlet scalar DM with LZ, invisible Higgs decay and gamma-ray line observations" [36]: the most recent (2024) combined analysis with LZ data.
- "GeV-scale thermal DM from dark photons" [31]: while focused on the vector portal, includes useful comparison of the Higgs-portal interplay in the dark Abelian Higgs model.

4 $U(1)_{B-L}$

The Z'_{B-L} couples to *all* SM fermions with charges proportional to $B-L$; anomaly cancellation requires three right-handed neutrinos, naturally incorporating the seesaw mechanism. Because the Z' has unsuppressed vector couplings to both quarks and leptons, the model is tightly constrained from multiple directions.

4.1 Direct detection

- **LZ (2025):** The LUX-ZEPLIN experiment sets the world-leading spin-independent (SI) cross-section limit of $\sigma_{\rm SI} \sim 9.2 \times 10^{-48}$ cm^2 at $m_{\rm DM} \sim 30$ GeV. For Z'-portal Dirac DM this excludes large regions with $g' \gtrsim 10^{-2}$ and $m_{Z'} \in [10, 150]$ GeV [1].
- **Light Z'_{B-L} DM:** A comprehensive study of light Dirac DM with a light Z'_{B-L} finds that SI Z'-exchange gives the strongest bound for $m_{Z'} \gtrsim 10$ GeV [2].
- **Conversion-driven DM in $U(1)_{B-L}$:** In the coscattering/conversion-driven freeze-out scenario, direct detection excludes $m_{Z'} \in [10, 150]$ GeV at $g' = 10^{-2}$, but the viable parameter space shifts to $g' \sim 3 \times 10^{-6}$–2×10^{-4}, $m_{Z'} \sim 0.02$–10 GeV—within reach of Belle II, FASER, and SHiP [3].

4.2 Collider constraints

- **LHC dilepton:** ATLAS/CMS dilepton resonance searches exclude $m_{Z'} \lesssim 5$ TeV for $\mathcal{O}(1)$ gauge couplings. For weaker couplings, dijet searches provide complementary coverage [4].
- **LEP / EW precision:** Z–Z' mixing is constrained by Z-pole observables (invisible width, forward-backward asymmetries).

4.3 Cosmological and astrophysical constraints

- **BBN / $N_{\rm eff}$:** A light Z'_{B-L} that thermalises with neutrinos contributes to $N_{\rm eff}$. Combined CMB + BBN data require $m_{Z'} \gtrsim 10$ MeV for $g' \sim 10^{-4}$ or larger. The "dilution-resistant" effect means even late-decoupling mediators leave an observable imprint [5].
- **Neutrino masses:** The $B-L$ model naturally incorporates the type-I seesaw mechanism via the right-handed neutrinos required for anomaly cancellation, linking DM phenomenology to the neutrino mass scale.

4.4 Key references

- PDG Dark Matter review (2024) [6]: general context for all WIMP models.
- 2024 TASI DM Primer [7]: includes Z'-portal models with $B - L$ as a benchmark.
- Conversion-driven DM in $U(1)_{B-L}$ [3]: closest to a comprehensive constraint compilation for this model.

5 $U(1)_{L_i - L_j}$ (Leptophilic Portals)

These extensions are anomaly-free without exotic fermions. The Z' couples to leptons only at tree level; coupling to quarks and nucleons arises at one loop via kinetic mixing with the photon, making direct detection naturally suppressed.

5.1 $L_\mu - L_\tau$

This is the most studied of the three $L_i - L_j$ models, historically motivated by the muon $g-2$ anomaly and neutrino mixing patterns.

Direct detection. Loop-induced Z'–photon mixing generates a tiny effective coupling to electrons and nucleons. Current direct-detection experiments (LZ, XENONnT, PandaX) are *not* competitive with collider and astrophysical bounds for this model [8].

Neutrino trident production. The $Z'_{\mu\tau}$ enhances $\nu_\mu N \to \nu_\mu \mu^+ \mu^- N$ via constructive interference with the SM Z/W. Combined CCFR, CHARM-II, and NuTeV data yield $\sigma_{\rm exp}/\sigma_{\rm SM}|_{\rm trident} = 0.93 \pm 0.25$, which is the strongest constraint for $m_{Z'} \gtrsim 60$ MeV [9].

Fixed-target and B-factory experiments.

- **NA64-μ (2024):** Covers large portions of the low-mass parameter space, strongly pressuring the $(g-2)_\mu$-favoured region [10].
- **BaBar 4μ:** Excludes $m_{Z'} \sim 0.2$–10 GeV for moderate couplings [11].
- **Future:** Belle II, MuSIC, FCC-ee, and LDMX will probe most of the remaining viable parameter space [12].

Astrophysical constraints.

- **White dwarf cooling:** Loop-induced Z'–photon mixing enables anomalous energy loss in hot white dwarfs via plasmon decay; constrains very light Z' [13].
- **SN1987A:** Light Z' production in the proto-neutron star constrains couplings in the MeV-mass regime [14].

Cosmological constraints. Very light Z' ($m \lesssim 10$ MeV) thermalises with neutrinos and contributes to $N_{\rm eff}$; detailed bounds are given in Ref. [15].

Key references.

- "Enabling Thermal DM within the Vanilla L_μ–L_τ Model" [12]: thorough 2025 analysis of the full constraint landscape (trident, NA64-μ, BaBar, BBN, WD cooling, relic density).
- "Global Neutrino Constraints on the Minimal $U(1)_{L_\mu-L_\tau}$ Model" [16]: combined oscillation, trident, and coherent scattering data (2025).
- "Light thermal DM in the light of DAMIC-M 2025 constraints" [17]: sub-GeV DM with L_μ–L_τ; DAMIC-M electron-scattering limits now exclude large portions of parameter space.

5.2 $L_e - L_\mu$ and $L_e - L_\tau$

These models differ from L_μ–L_τ in that the Z' couples to electrons at tree level, significantly changing the phenomenology:

- Direct detection via DM–electron scattering is unsuppressed (for sub-GeV DM) or enhanced relative to the L_μ–L_τ case.
- Planetary perihelion precession constrains ultra-light Z' mediating a long-range e–e force [18].
- $(g-2)_e$ provides direct sensitivity to the L_e-coupled Z'.

Comparative analysis. The most useful single reference covering all three L_i–L_j models is Ref. [19], which analyses leptophilic DM for the Fermi-LAT Galactic Centre Excess (GCE):

- L_μ–L_e: best fit to the GCE with $m_{\mathrm{DM}} \sim 40$–50 GeV, $m_{Z'} \sim 70$–86 GeV. Consistent with $(g-2)_{e,\mu}$, collider, and direct detection bounds.
- L_e–L_τ and L_μ–L_τ: poorer GCE fits; experimental constraints force the DM mass near resonance ($m_{\mathrm{DM}} \approx m_{Z'}/2$).
- All three evade current direct-detection bounds due to their leptophilic nature.
- CMB (Planck): s-wave annihilation constrained for $m_{\mathrm{DM}} \lesssim 10$ GeV.

Global fit to Z' couplings. The "Global Analysis of Leptophilic Z' Bosons" [20] performs a comprehensive fit using $(g-2)_{e,\mu}$, lepton-flavour-violating decays ($\mu \to 3e$, $\tau \to 3\mu$), Z-pole observables, and $\tau \to \mu\nu\bar{\nu}$ universality tests. This is the best single reference for the full constraint landscape on the Z' couplings themselves, independent of the DM sector.

6 $U(1)_B$ (Gauged Baryon Number / Baryon Portal)

The Z'_B is *leptophobic*—it couples to quarks but not to leptons at tree level. This makes it invisible in dilepton searches, but direct detection via Z'–nucleon scattering is generically unsuppressed.

6.1 Direct detection

- **Anomaly-free classification (Duerr, Kahlhoefer et al.):** Ref. [21] shows that anomaly cancellation in $U(1)_B$ admits exactly four minimal dark-sector solutions in which the DM–Z' coupling is *axial*. This makes SI scattering momentum-suppressed ($\sigma_{\mathrm{SI}} \propto q^4$), while spin-dependent cross sections remain but are experimentally weaker. The scenario is viable under LZ/PandaX bounds.

- **Singlet-doublet fermionic DM (2025):** Ref. [22] considers $U(1)_B$ with anomaly cancellation via three exotic fermions; a residual $\mathbb{Z}_2$ stabilises the DM. Two-component singlet-doublet DM relaxes SI bounds through suppressed mixing. Higgs-portal exchange constrains the Higgs mixing angle to $\lesssim 0.22$.
- **Asymptotically safe DM:** Ref. [23] studies TeV-scale Majorana DM with gauged baryon number in the asymptotic-safety framework; kinetic mixing becomes a derived quantity. Consistent with PandaX-II for $m_{\mathrm{DM}} \gtrsim 1$ TeV.

6.2 Collider constraints

- **LHC dijet resonance searches:** The primary discovery channel for a leptophobic Z'. Current ATLAS/CMS bounds exclude $m_{Z'_B} \lesssim 500$ GeV to multi-TeV, depending on the gauge coupling [24].
- **Monojet:** Complementary for lighter Z'_B or compressed mass spectra [25].
- **No dilepton signal:** Unlike $B - L$, the leptophobic nature means dilepton searches provide no constraint at tree level.

6.3 Cosmological and astrophysical constraints

- **BBN:** Less constraining than $B - L$ since Z'_B does not couple to neutrinos at tree level.
- **SN1987A:** Z'_B produced via nucleon bremsstrahlung in the proto-neutron star; constrains light Z'_B ($m \lesssim 200$ MeV) for moderate couplings [14].
- **Gravitational wave signatures:** A first-order $U(1)_B$ phase transition could produce a stochastic GW background detectable by LISA/Einstein Telescope [26].

6.4 Key references

- Duerr, Kahlhoefer et al. (2018) [21]: classification of anomaly-free solutions and direct-detection phenomenology.
- Singlet-doublet DM in gauge theory of baryons (2025) [22]: most recent comprehensive study.
- Asymptotically safe DM with gauged baryon number [23]: TeV-scale constraints.

7 Summary comparison

References

[1] LZ Collaboration, "Dark Matter Search Results from 417 Live Days of LUX-ZEPLIN (LZ) Data," (2025). LBL news release; Phys. Rev. Lett.

[2] D. Lamprea and J. Zurita, "Light Z' and Dirac fermion dark matter in the $B - L$ model," Eur. Phys. J. C **82**, 901 (2022), doi:10.1140/epjc/s10052-022-10801-3.

[3] "Conversion-Driven Dark Matter in $U(1)_{B-L}$," (2024), arXiv:2411.06744.

[4] F. Kahlhoefer, K. Schmidt-Hoberg, T. Schwetz, and S. Vogl, "Constraints on Z' models from LHC dijet searches," (2016), arXiv:1605.07940.

[5] "N_{eff} constraints on light mediators coupled to neutrinos: the dilution-resistant effect," JHEP **10**, 012 (2023), doi:10.1007/JHEP10(2023)012.

[6] PDG Dark Matter review (2024), pdg.lbl.gov.

[7] "2024 TASI Lectures: A Dark Matter Primer," arXiv:2506.05234.

[8] "Direct detection of light dark matter charged under a $L_\mu - L_\tau$ symmetry," (2024), arXiv:2404.03090.

[9] W. Altmannshofer, S. Gori, M. Pospelov, and I. Yavin, "Neutrino Trident Production: A Powerful Probe of New Physics with Neutrino Beams," Phys. Rev. Lett. **113**, 091801 (2014), arXiv:1406.2332.

[10] NA64 Collaboration, "First constraints on the $L_\mu - L_\tau$ explanation of the muon $g-2$ anomaly from NA64-e at CERN," (2024), arXiv:2404.06982.

[11] BaBar Collaboration, 4μ search. See discussion in Ref. [12].

[12] "Enabling Thermal Dark Matter within the Vanilla L_μ–L_τ Model," Phys. Rev. D (2025), arXiv:2507.02048.

[13] "White dwarf cooling through neutrinos and $L_\mu - L_\tau$," talk at Invisibles24 Workshop (2024), Indico.

[14] "Cooling the Shock: New Supernova Constraints on Dark Photons," Phys. Rev. Lett. **134**, 151002 (2025), doi:10.1103/PhysRevLett.134.151002.

[15] M. Escudero, D. Hooper, G. Krnjaic, and M. Pierre, "Cosmology With a Very Light $L_\mu - L_\tau$ Gauge Boson," JHEP **03**, 071 (2019), arXiv:1901.02010.

[16] "Global Neutrino Constraints on the Minimal $U(1)_{L_\mu - L_\tau}$ Model," Phys. Rev. D **111**, 095034 (2025), PRD.

[17] D. Borah et al., "Light thermal dark matter models in the light of DAMIC-M 2025 constraints," (2025), arXiv:2509.16319.

[18] "Constraints on long range force from perihelion precession of planets in a gauged $L_e - L_{\mu,\tau}$ scenario," Eur. Phys. J. C **81**, 286 (2021), doi:10.1140/epjc/s10052-021-09078-9.

[19] "Leptophilic dark matter in $U(1)_{L_i - L_j}$ models: a solution to the Fermi-LAT Galactic Center Excess," (2025), arXiv:2508.02775.

[20] A. Greljo, Y. Soreq, P. Stangl, A. E. Thomsen, and J. Zupan, "Global Analysis of Leptophilic Z' Bosons," JHEP **06**, 068 (2021), arXiv:2104.07680.

[21] M. Duerr, F. Kahlhoefer, K. Schmidt-Hoberg, T. Schwetz, and S. Vogl, "Anomaly-free Dark Matter with Harmless Direct Detection Constraints," JHEP **12**, 126 (2018), arXiv:1807.07921.

[22] "Singlet-doublet fermionic dark matter in gauge theory of baryons," JHEP **01**, 159 (2025), doi:10.1007/JHEP01(2025)159.

[23] "Asymptotically safe dark matter with gauged baryon number," Phys. Rev. D **107**, 035018 (2023), doi:10.1103/PhysRevD.107.035018.

[24] See Ref. [4] and references therein; ATLAS/CMS dijet resonance search summaries.

[25] "Constraining dark sectors with monojets and dijets," JHEP **07**, 089 (2015), doi:10.1007/JHEP07(2015)089.

[26] "Probing Leptophobic Dark Sectors via Gravitational Wave Signatures," (2025), arXiv:2508.17476.

[27] X. G. Wang, B. M. Loizos, and A. W. Thomas, "WIMP Dark Matter within the dark photon portal," (2025), arXiv:2508.08676.

[28] "Experimental targets for dark photon dark matter," Phys. Rev. D **111**, 103535 (2025), arXiv:2410.14774.

[29] "The Heavy Dark Photon Handbook: Cosmological and Astrophysical Bounds," (2025), arXiv:2511.15785.

[30] "Combined constraints on dark photons from high-energy collisions, cosmology, and astrophysics," (2026), arXiv:2601.15066.

[31] D. Alonso-González, D. Cerdeño, P. Foldenauer, and J. M. No, "GeV-scale thermal dark matter from dark photons: tightly constrained, yet allowed," (2025), arXiv:2507.11376.

[32] "Exploring Dark Photon Production and Kinetic Mixing Constraints in Heavy-Ion Collisions," Astron. Nachr. (2025), arXiv:2412.02536.

[33] "Dark Photon Searches with Initial-State Radiation at Fixed-Target Configurations," (2025), arXiv:2505.10302.

[34] "BBN Constraints on the Hadronic Annihilation of sub-GeV Dark Matter," (2025), arXiv:2510.11791. See also Ref. [5] and "Joint CMB and BBN Constraints on Light Dark Sectors with Dark Radiation," arXiv:2109.03246.

[35] "Revisiting dark photon constraints from CMB spectral distortions," MNRAS **535**, 1874 (2024), arXiv:2409.12115.

[36] "New constraints on singlet scalar dark matter model with LZ, invisible Higgs decay and gamma-ray line observations," (2024), arXiv:2410.21089.

[37] "Direct detection of Higgs portal for light self-interacting dark matter," Eur. Phys. J. C **85**, 317 (2025), arXiv:2410.20645.

[38] "Dark Matter through the Higgs portal," (2019), arXiv:1903.03616.

[39] GAMBIT Collaboration, "Global analyses of Higgs portal singlet dark matter models using GAMBIT," Eur. Phys. J. C **79**, 38 (2019), doi:10.1140/epjc/s10052-018-6513-6.

[40] "LHC-friendly freeze-in dark matter via the Higgs portal," Phys. Rev. D **112**, 035034 (2025), ADS.

[41] "Light thermal dark matter beyond p-wave annihilation in minimal Higgs portal model," JHEP **05**, 281 (2024), doi:10.1007/JHEP05(2024)281.

[42] "Revisiting Singlet Fermion Dark Matter with a Scalar Portal: Connecting Higgs Phenomenology and Strong Electroweak Phase Transition," (2026), arXiv:2601.13147.

[43] Fermi-LAT Collaboration, "Legacy analysis of dark matter annihilation from the Milky Way dwarf spheroidal galaxies with 14 years of Fermi-LAT data," (2024), ResearchGate.

[44] "Gamma-ray lines in 15 years of Fermi-LAT data: New constraints on Higgs portal dark matter," Phys. Rev. D **109**, L041301 (2024), doi:10.1103/PhysRevD.109.L041301.

Constraint	**Vector (dark photon)**	**Higgs portal**	$\boldsymbol{B-L}$	$\boldsymbol{L_i-L_j}$	$\boldsymbol{U(1)_B}$
SI direct detection	Strong ($\propto\varepsilon^2$)	Strong ($\propto\sin^2\theta$)	Strong (tree)	Weak (loop)	Suppressed if axial
LHC dilepton	Moderate	—	Very strong	Weak (leptophilic)	None at tree level
LHC dijet/monojet	Moderate	Moderate (VBF, mono-Z)	Moderate	None	Strong
Invisible Higgs	—	Strong ($m_S<m_h/2$)	—	—	—
ν trident	—	—	—	Strong (L_μ–L_τ)	—
BaBar / Belle II	Strong	Moderate	Moderate	Strong	Weak
Fixed target (NA64, LDMX)	Very strong	Some reach	Some reach	Strong (NA64-μ)	Some reach
BBN / $N_{\rm eff}$	Strong (light A')	Strong (long-lived S)	Strong (light Z')	Strong (light Z')	Weak
Stellar cooling (WD, SN)	Very strong	Moderate	Moderate	Moderate (via loop mixing)	SN for light Z'
Indirect (γ-ray)	Moderate	Strong (dwarfs, lines)	Moderate	Moderate	Weak
CMB (Planck)	Strong (s-wave)	Moderate (p-wave)	Moderate	Moderate	Weak

Table 1: Qualitative comparison of the leading constraints across the five portal models.